Experimental Design: 2 × 2 × 2 Factorial + Apparatus

38 Business Problems × 5 Model Families × 10 Replicates = 1,900 Runs

TIER	STRUCTURE	PERSONA
Cheap (C)	Solo (S)	None (N)
Frontier (F)	Team (T)	Loaded (P)

crossed

C-S-N n=285	**C-S-P** n=285	**C-T-N** n=285	**C-T-P** n=285
F-S-N n=190	**F-S-P** n=190	**F-T-N** n=190	**F-T-P** n=190

Study 2: refinement

Generator(s)

builder → critic → refiner

answer

Judge Panel (blinded)

3 judges × 3 prompt variants